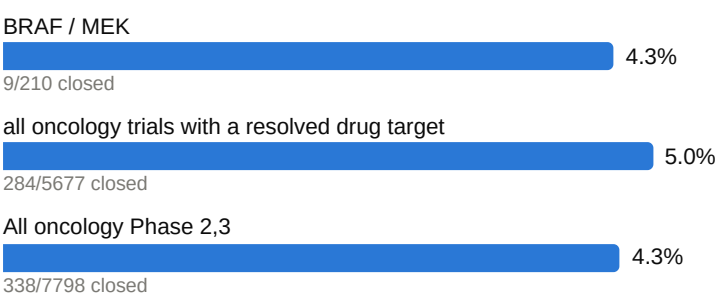


# BRAF / MEK: 7 molecules stopped early, all small molecules

9 stopped trials, but 7 molecules — all small molecules: Cobimetinib, Regorafenib, Trametinib, Binimetinib and 3 more. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

|                                           |                                      |                                                          |                                                                                                                  |
|-------------------------------------------|--------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 4.3%                                      | 3 / 1 / 1                            | 43.8%                                                    | 1.7%                                                                                                             |
| 9 of 210 closed trials (95% CI 2.3%–7.9%) | efficacy / safety / both (adds to 9) | of 480 trials have closed · 270 still open or unresolved | stopped for a biological reason within 3 years of starting (95% CI 0.8%–3.7%; 255 still at risk; 1.5% area-wide) |

## How this compares



**What is in this class:** 277 distinct experimental drugs across 480 trials (small molecule 480, antibody 178, unknown 61, adc 16). Most tested: REGORAFENIB (109), TRAMETINIB (85), SORAFENIB (76), BINIMETINIB (65), COBIMETINIB (60). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 9 stops came from 9 sponsor-asset programmes across 9 sponsors; the largest (UNICANCER) contributed 1; removing that programme's trials from both sides leaves 3.8%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction ( $q=1$ , Benjamini-Yekutieli). A further 11 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 9.5%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI     |
|----------------|-------|--------|------|------------|
| 2015–2017      | 2     | 93     | 2.2% | 0.6%–7.5%  |
| 2018–2020      | 3     | 72     | 4.2% | 1.4%–11.5% |
| 2021–2024      | 4     | 45     | 8.9% | 3.5%–20.7% |

## Where the stops sit

Benefit–risk: 4, Efficacy / futility: 3, Safety: 1, Efficacy and safety: 1. Sponsors with most stops: Cancer Trials Ireland (1), Great Novel Therapeutics Biotech & Medicals Corporation (1), SpringWorks Therapeutics, Inc., a healthcare company of Merck KGaA, Darmstadt, Germany (1), Bristol Myers Squibb (1), UNICANCER (1), Columbia University (1).

## The stopped trials

| Trial       | Ph | Sponsor                                                 | Experimental drugs             | Stopped | Type         | Registry stop reason                                                          |
|-------------|----|---------------------------------------------------------|--------------------------------|---------|--------------|-------------------------------------------------------------------------------|
| NCT05518110 | 2  | Cancer Trials Ireland                                   | TRAMETINIB, HYDROXYCHLOROQUINE | 2026-03 | Efficacy     | Study terminated due to futility                                              |
| NCT05770882 | 2  | Great Novel Therapeutics Biotech & Medicals Corporation | TUCIDINOSTAT, REGORAFENIB      | 2025-01 | Benefit–risk | The overall profile does not support development for Hepatocellular Carcinoma |

| Trial       | Ph | Sponsor                                                                                                | Experimental drugs                            | Stopped | Type              | Registry stop reason                                                                                                      |
|-------------|----|--------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| NCT05580770 | 2  | SpringWorks Therapeutics, Inc., a healthcare company of Merck KGaA, Darmstadt, Germany                 | MIRDAMETINIB, BRIMARAFENIB                    | 2025-01 | Safety            | Lack of tolerability of BGB-3245 (Brimarafenib) when given in combination with Mirdametininb                              |
| NCT05071183 | 2  | Bristol Myers Squibb  | REPOTRECTINIB, TRAMETINIB                     | 2023-03 | Benefit–risk      | Inability to achieve and optimize a dose that could provide a positive benefit risk profile                               |
| NCT03722108 | 2  | UNICANCER                                                                                              | REGORAFENIB, IRINOTECAN                       | 2022-05 | Benefit–risk      | Unfavorable benefit-risk balance in the experimental arm following the IDMC for intermediate efficacy and safety analysis |
| NCT04214418 | 2  | Columbia University                                                                                    | COBIMETINIB, HYDROXYCHLOROQUINE, ATEZOLIZUMAB | 2022-04 | Efficacy + safety | Toxicity and Lack of Efficacy                                                                                             |

The 6 most recent of 9 stopped trials. Every molecule behind all 9 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

**Method**

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).  
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 43.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**Any mechanism, sponsor or indication, updated weekly.** The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: [clinicaltrialfailures.com/data-licensing](https://clinicaltrialfailures.com/data-licensing)

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.